Lesson 12 Teacher's Checklist / Prep Guide

Complete before class

VIDEO GAME CREATION

A New Phase Begins

Lesson 11 reached Level 2 Complete - the mechanics are done. From here through Lesson 15, every lesson is about making that same loop feel good: sound this week, animation in Lesson 13, difficulty scaling in Lesson 14, and a final polish pass in Lesson 15. This is a lighter, more satisfying lesson than most - little new mechanical difficulty, mostly wiring up cues students can immediately hear.

This Week: The Sound Payoff

Since Lesson 2, several lessons have quietly offered an optional 'add a sound' Level 3 idea: a movement blip (Lesson 2), a shoot sound (Lesson 3), a hit/explosion sound (Lesson 4), and a victory sound (Lesson 6). Lesson 8 made life-lost and game-over sounds optional as well. Some students built these; many didn't. This week makes every one of those sounds mandatory for everyone - shoot, explosion, win, life-lost, game-over - plus one genuinely new sound (wave clear), matching the capstone's Level 3 feature list exactly. For students who already built the optional versions, today is mostly verification plus one new sound. For everyone else, this lesson gives full instructions as if starting fresh, the same self-contained approach every lesson uses.

- ☐ Read the Handout's Setup and Level 1 sections before class - for students who skipped the earlier optional sound ideas, this is effectively new material, not just a quick add-on.

Self-Contained Lesson Assets

One new sound effect is provided in this lesson's Assets folder: sfx_wave.wav (a short, upbeat wave-clear cue) - distinct from the existing sfx_shoot.wav, sfx_hit.wav, sfx_win.wav, sfx_lifelost.wav, and sfx_gameover.wav so all six are easy to tell apart by ear. The game-over sound reuses Lesson 8's existing sfx_gameover.wav asset rather than introducing a new one - no separate 'lose' sound exists in this course.

- ☐ Confirm sfx_wave.wav is present and playable before class (a quick double-click test on the demo laptop is enough).

Continuity From Lesson 11

- ☐ Remind students to open their own [YourName]Game project from Lesson 11, not create a new one.

- ☐ If a student's Lesson 11 project is lost or broken, have a pre-built "Lesson 11 complete" starter available per engine (from the Lesson 11 Teacher's Keys) so they aren't blocked.

Engine Verification

- ☐ Confirm the demo laptop's Lesson 10-11 exercises (restart flow, multiple waves) still work for all 8 engines.
- ☐ Test a full round on every engine before class: a shot, a hit, an early wave clear, and a loss should each play a distinct, correctly-timed sound with no missing or doubled playback.
- ☐ Specifically test that the wave-clear sound and the win sound never both play on the same frame (the true-win branch and the wave-advance branch are mutually exclusive, but it's an easy mistake to place a sound call outside the if/else split).
- ☐ 001 Game Creator and Unreal (Windows demo laptop): confirm newly imported sfx_wave.wav is recognized correctly by each engine's audio import step.

Materials

- ☐ Print or queue digital copies of all 8 Student Handouts (or confirm students can access their engine's version electronically).
- ☐ Have the 8 Teacher's Keys on hand (instructor reference only - do not distribute) in case a student needs a worked example beyond Level 1.
- ☐ Bring: demo laptop, projector cable/adapter, handout copies, and headphones or speakers for the demo (this is the first lesson where audio output actually matters for the lecture demo).

Known Gotchas This Week

- Students who already built an optional sound idea in an earlier lesson may assume today needs nothing from them - point out explicitly that today still adds one brand-new sound (wave clear) and makes the game-over/life-lost sounds mandatory if they skipped them.
- The most likely mistake: placing a sound call outside the if/else split from Lesson 11, causing two round-ending sounds to play at once, or the wrong one to play.
- A close second: forgetting that Scratch's "start sound" and most other engines' one-shot sound calls don't block gameplay - this is normal and expected, not a bug to fix.
- Encourage a genuinely playful few minutes of testing this week - hearing a game respond to your actions is one of the most satisfying moments in the whole course, and it's worth letting students enjoy it.